

Assessment of the `carryover_focus/` Directory

pmsimstats2025 Project Notes

March 2025

1 Summary

This directory is a self-contained sub-analysis that distills the 144-condition full simulation (`simulation_clustered`) down to 9 conditions, isolating a single question: what happens to statistical power for biomarker moderation detection when pharmacological carryover is present but the analysis model ignores it?

2 Contents

File	Role
<code>PLAN.md</code>	Design document with line-level extraction references to the parent script
<code>README.md</code>	User-facing documentation (how to run, expected results)
<code>carryover_power_simulation.R</code>	701-line self-contained simulation (functions extracted verbatim from <code>simulation_clustered.R</code>)
<code>predictor_comparison.tex/.pdf</code>	7-page technical note documenting a key finding that emerged during this work
<code>output/</code>	Three <code>.RData</code> result sets and corresponding figures for different <code>tsd_min</code> values

3 Intellectual Arc

The work progressed through three distinct phases.

3.1 Phase 1: Simplification

The full parameter grid in `simulation_clustered.R` sweeps $4 \text{ designs} \times 2 \text{ sample sizes} \times 4 \text{ biomarker moderation levels} \times 3 \text{ carryover values} \times 2 \text{ model specifications} = 144 \text{ conditions}$. This sub-analysis reduces the grid to $\text{carryover half-life} \times \text{model specification}$ for the Hybrid design alone.

A new predictor, `naive_drug_weeks`, was introduced as the “ignores carryover” comparator. It replaces the original `weeks_on_drug`, which was structurally deficient: because `weeks_on_drug = cumsum(treatment)` never decreases during off-drug periods, it provides no on/off contrast and yields power ≈ 0.10 regardless of the carryover level. The replacement `naive_drug_weeks = if_else(treatment == 1, weeks_on_drug, 0)` drops to zero when off drug, preserving the cumulative scale while restoring the contrast needed for the interaction test.

3.2 Phase 2: Discovery of the $t_{sd} = 0$ Discontinuity

The initial results (the `tsd0` output set) showed a step-function pattern: power drops from ~ 0.81 at $t_{1/2} = 0$ to ~ 0.20 at $t_{1/2} = 0.05$ weeks (8.4 hours), then remains flat regardless of further increases in half-life.

The `predictor_comparison.tex` document traces this to a mathematical artifact. At the exact moment of discontinuation, $t_{sd} = 0$, and the carryover function evaluates as:

$$\left(\frac{1}{2}\right)^{0/t_{1/2}} = \left(\frac{1}{2}\right)^0 = 1 \quad \text{for any } t_{1/2} > 0$$

Since weeks 10 and 20 in the Hybrid design have $t_{sd} = 0$, they carry full drug contamination regardless of whether the half-life is 8 hours or 34 hours. These two timepoints constitute half of the off-drug measurements, producing severe and constant attenuation of the on/off contrast. This same artifact is present in Hendrickson et al.’s original `pmsimstats` code.

3.3 Phase 3: The $t_{sd,min}$ Correction

Setting `tsd_min` = 0.5 weeks imposes a realistic gap between the last dose and the first off-drug measurement, restoring continuity in the carryover function:

$$t_{sd,eff} = \max(t_{sd}, t_{sd,min})$$

The corrected simulation (the `tsd05` output set) produces the expected gradual, monotonic power decline as the true carryover half-life increases.

4 The Three Output Sets

Table 1: Output sets generated by `carryover_power_simulation.R` under different $t_{sd,min}$ values.

Suffix	$t_{sd,min}$	Pattern
<code>_tsd0</code>	0 (default)	Step-function cliff. Modeled power ~ 0.80 across all half-lives; naive power drops to ~ 0.20 at any nonzero half-life and stays flat. This is the discontinuity artifact.
<code>_tsd05</code>	0.5 weeks	Gradual decline. Both lines start at ~ 0.80 at carryover = 0; the naive line declines smoothly as the half-life increases while the modeled line holds steady. This is the pharmacologically expected pattern.
(no suffix)	0.5 weeks	Appears identical to the <code>tsd05</code> set based on the figure, suggesting the most recent run used $t_{sd,min} = 0.5$.

5 Relationship to the Broader Project

This sub-analysis serves three purposes for the paper:

1. **Pedagogical figure.** The two-panel power/effect-size line plot is far more readable than the full simulation’s heatmaps for communicating the carryover finding to a clinical audience.
2. **Identifies an artifact in Hendrickson et al. (2020).** The $t_{sd} = 0$ discontinuity explains why the published simulation heatmaps show an implausibly sharp power cliff at tiny carryover half-lives. The artifact affects all trial designs that place measurements at the treatment boundary.
3. **Proposes a correction.** The $t_{sd,min}$ parameter eliminates the mathematical discontinuity and produces pharmacologically realistic power curves. The recommendation is to report both the uncorrected (sharp cliff) and corrected (gradual decline) results, demonstrating that the power-cliff finding is partly an artifact of the $t_{sd} = 0$ assumption.

6 Open Items

- Per `PLAN.md`, the “run and verify” and “compare against `simulation_clustered_results.RData`” steps remain unchecked, although the output artifacts exist. Formal verification against the parent simulation’s results would confirm numerical consistency between the simplified and full codebases.
- The `PLAN.md` checklist has not been updated to reflect the completed work (script creation, README, output generation).

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Source: `~prj/alz/01-pmsimstats/pmsimstats2025/analysis/scripts/carryover_focus/directory_assessment.tex`